

# Report 9: Histogram Equalization

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## 1 Implementation

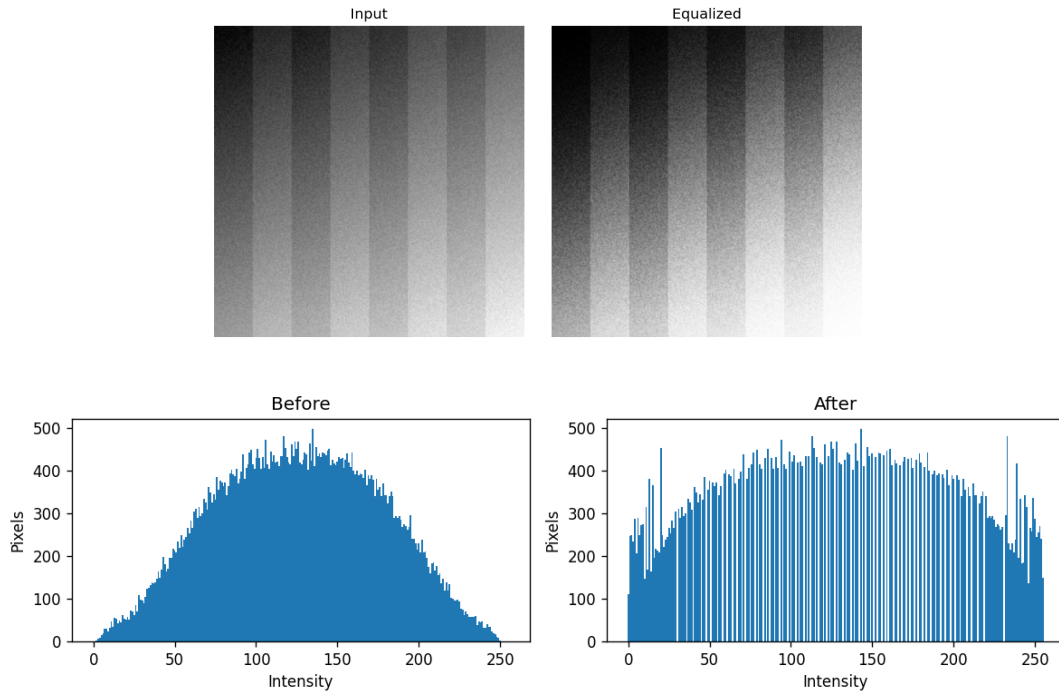
I used atomic additions to count 256 grayscale histogram bins. The histogram is reset before each measured run. On the CPU, I calculated a CDF lookup table; a  $16 \times 16$  CUDA kernel applies it to the image. Constant images are preserved. I checked the histogram and output against NumPy.

I ran the notebook on Google Colab with a Tesla T4 and a  $256 \times 256$  sample image. GPU timings use `time.time()` after warm-up and synchronization, averaged over three runs; allocation and image transfers are excluded. Where measured, CPU time is one run; speedup is relative to that Python CPU implementation.

## 2 Results

| Operation        | GPU (ms) |
|------------------|----------|
| Histogram        | 0.1205   |
| Equalization map | 0.0927   |

Histogram count: 65536 pixels, equal to  $256 \times 256$ . Maximum output error: 0. CDF construction is performed on the CPU and is not included in the kernel times.



## 3 Discussion

The histogram and mapped output matched the NumPy reference. The before/after histograms show how equalization redistributes grayscale intensities.